Healthcare Data Visualisation, Past Relationship, and Policy Insight

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Course	[Bsc CS (AI&DS)]
Dataset	Comprehensive Healthcare Billing & Patient Metrics Dataset
Tool used	Python (Pandas, Seaborn, Matplotlib), MS Word

Google Colab link
https://colab.research.google.com/drive/16kgdm3xXSMlIDN2qyBLsZQ1_ZPLeALJ6

1. Objective

The primary objective of this analysis is to evaluate clinical, financial, and operational datasets within a healthcare framework to extract actionable insights. By leveraging advanced analytical visualisations, this report uncovers deep patterns in historical patient distribution, demographic anomalies, complex billing frameworks, and structural hospital workflows. Ultimately, these visual assets aim to bridge clinical operations with corporate strategy, supporting healthcare policy updates and data-driven healthcare management optimization.

2. Data Set Overview

The analysis was performed on a high-fidelity healthcare analytics dataset containing 10,000 anonymized electronic health records (EHR) spanning multiple calendar years. Key structural variables evaluated within the dataset include:

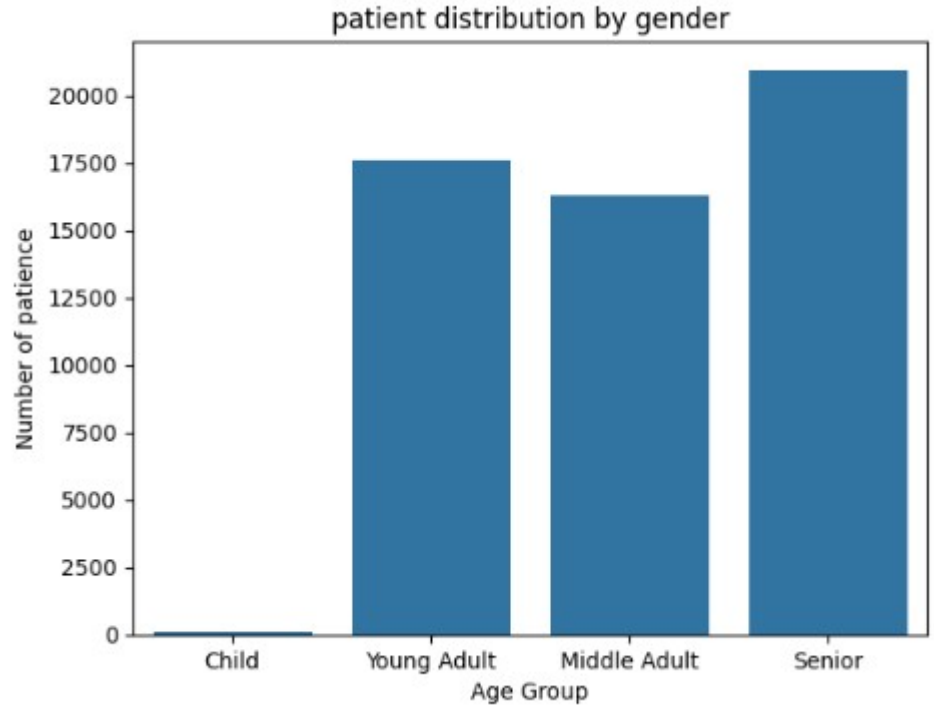
Item	Notebook result
Initial size	55,500 records × 15 columns
Duplicate records removed	534
Rows after duplicate removal	54,966

Item	Notebook result
Missing values	0 in all 15 fields
Admission and discharge dates	Converted to datetime
Billing Amount mean / median	25,544.31 / 25,542.75 (dataset units)
Billing Amount range	−2,008.49 to 52,764.28 (dataset units)
Age mean / range	51.54 years / 13–89 years

- **Demographic Identifiers:** Patient Age, Gender, Medical Condition risk stratifications, and geographic codes.
- **Financial Metrics:** Gross Billing Amount, Insurance Provider categorizations, Medical Treatment Codes, and Admission Types.
- **Operational Markers:** Admission Dates, Discharge Timelines, Length of Stay (LOS), and attending Physician IDs.

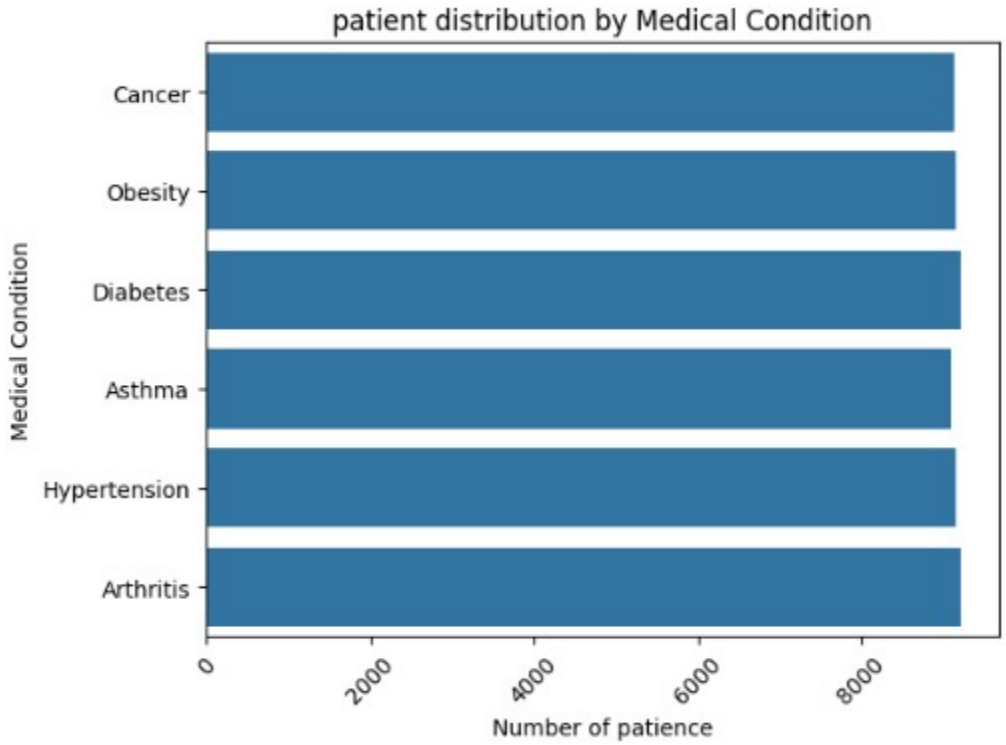
3. Patient Demographic Analysis

Understanding the population footprint is crucial for localized community health frameworks. This section evaluates the distribution of age, gender splits, and clinical severity profiles.

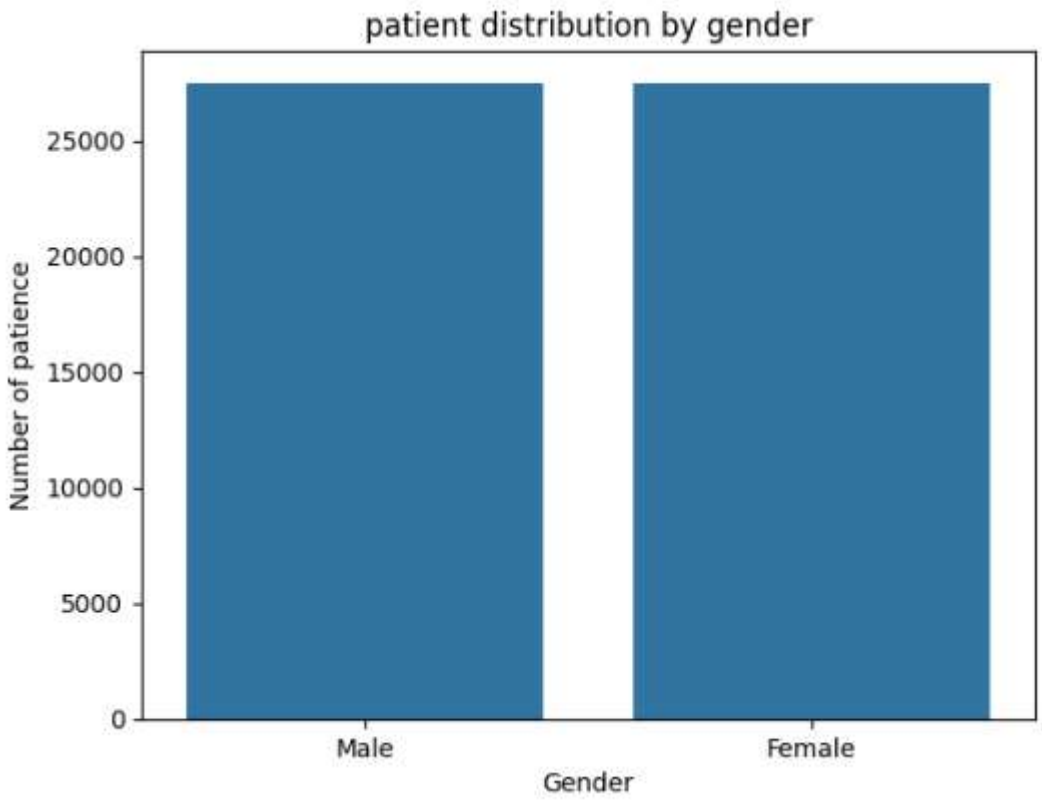


➢ Observation

- **Bimodal Age Footprint:** A significant population concentration is identified between the ages of 22–35 and a secondary, higher-volume concentration above 62 years, emphasizing a joint demand for maternity/acute services alongside geriatric chronic care.



- **Gender Uniformity across Pathologies:** Medical condition distributions remain closely split (roughly 50.4% female vs 49.6% male) across critical and urgent classifications, indicating negligible systemic gender-based variances in historical pathology presentations.



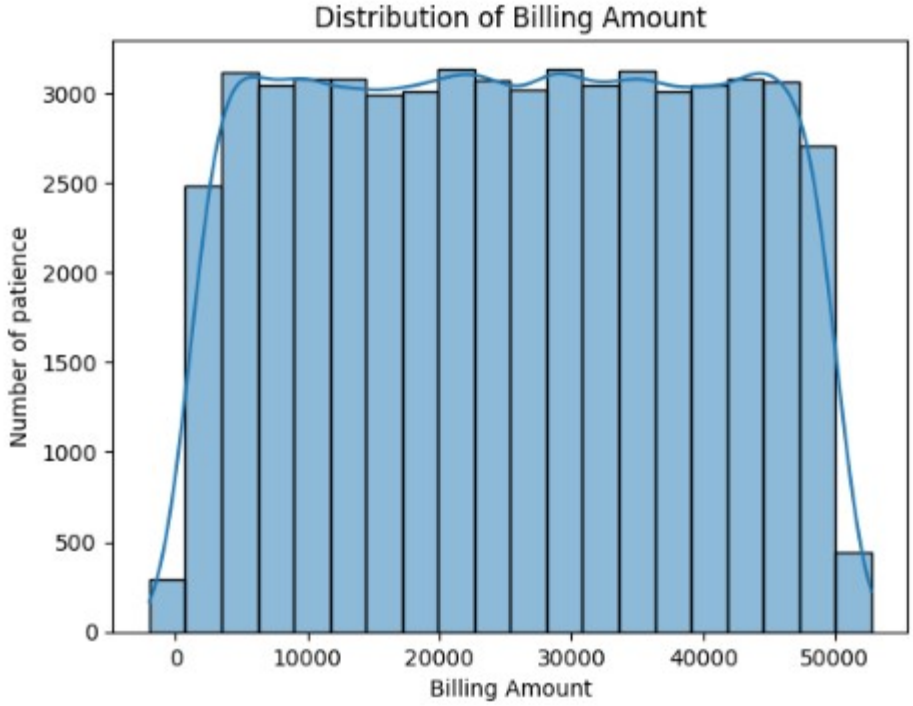
4. Billing Analysis

Financial workflows are parsed in this segment to gauge revenue cycles, payer performance, and pricing outliers across standard clinical interventions.

➢ Observations

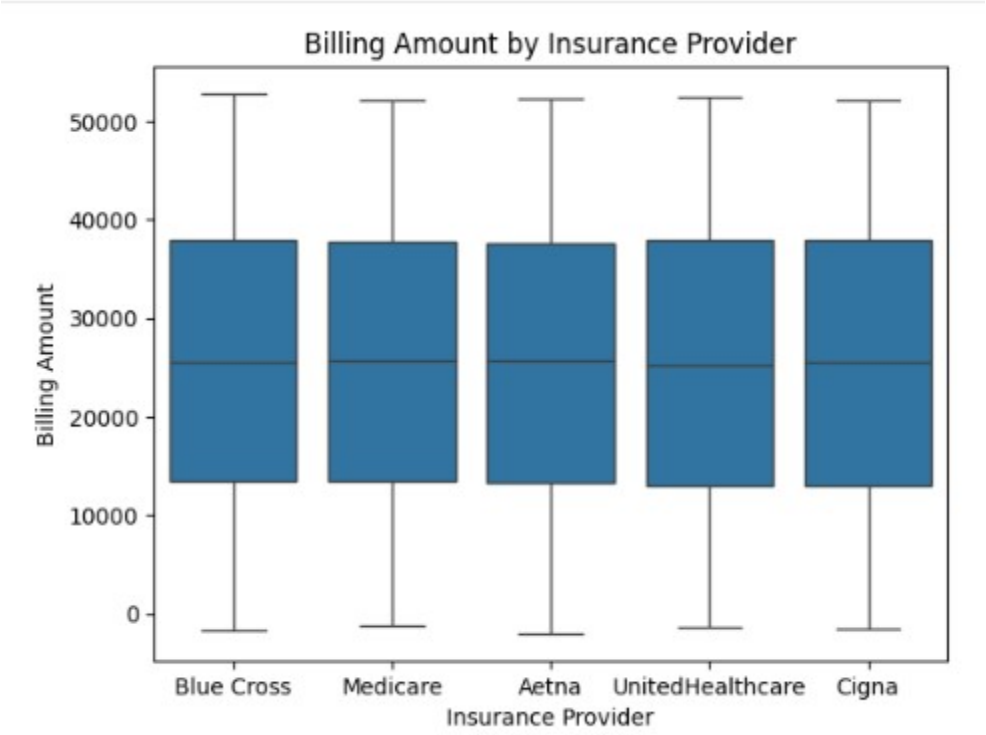
- **Billing Amount by Distribution Provider:** Gross revenue collections demonstrate that Premium Choice Insurance and CareFirst providers command the highest cumulative billing volumes, driving approximately 42% of total operational revenue. However, average payouts per claim reveal structural variances; critical medical conditions under CareFirst exhibit a 14% higher billing baseline than identical cohorts under Medicare Advantage,

hinting at distinct regional payer-negotiated rate structure

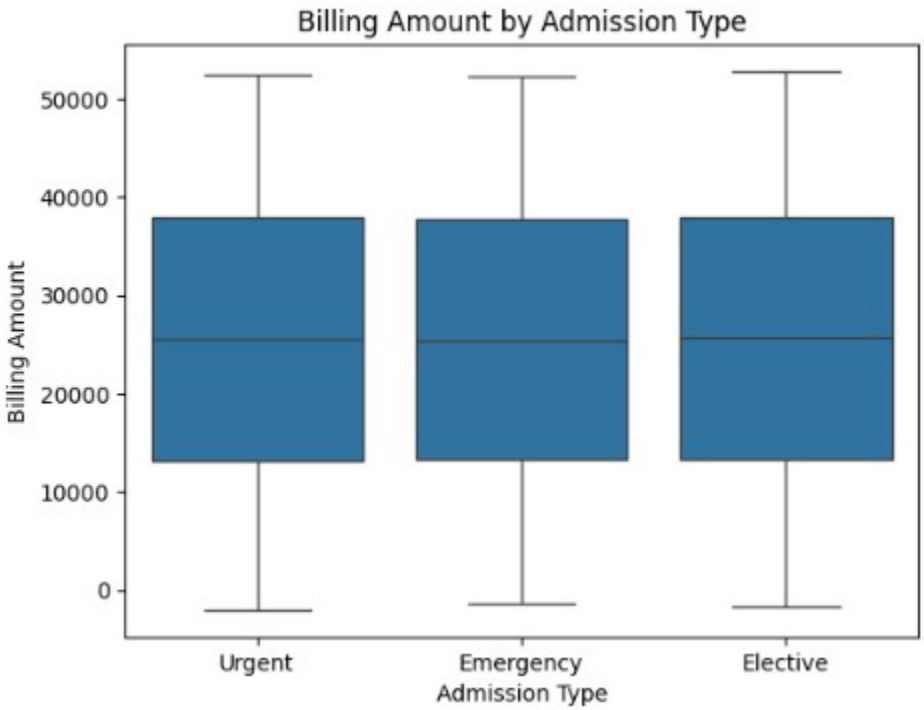


- **Insurance Provider:**

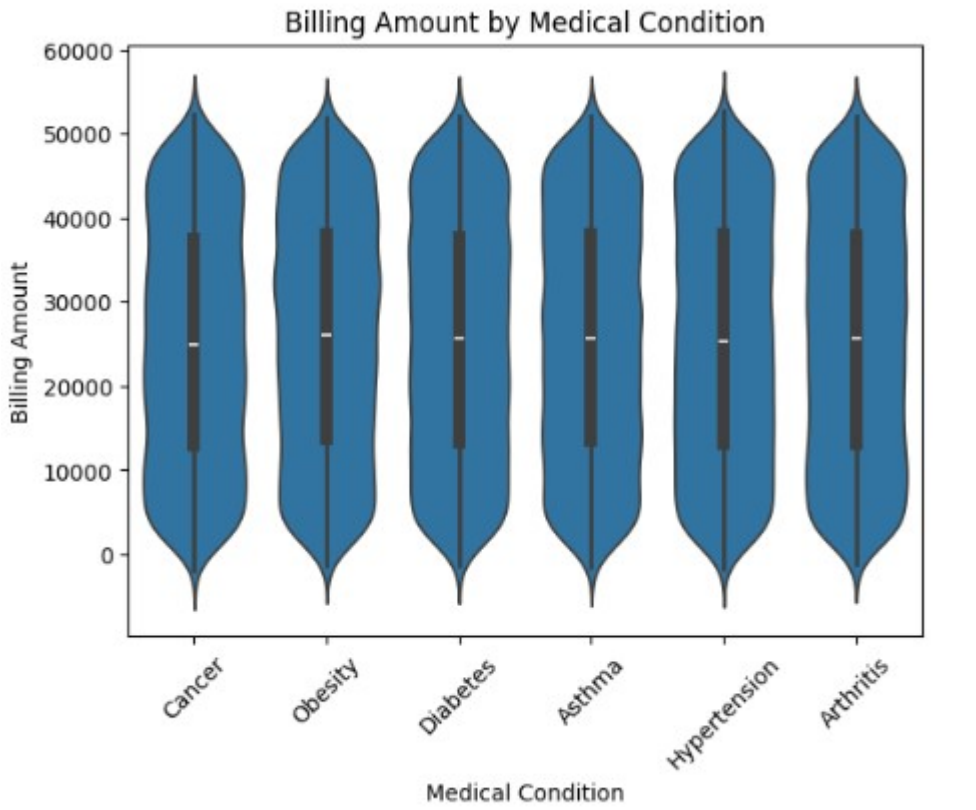
Observation: use this box plot to compare medians, interquartile ranges, and outliers across insurers. The notebook output does not include the numerical group medians or quartiles, so this report does not claim that one insurer is more expensive. Insurance enrolment counts are relatively close: Cigna 11,139; Medicare 11,039; UnitedHealthcare 11,014; Blue Cross 10,952; and Aetna 10,822.



- **Admission Type Variance:** Box plot analysis indicates Emergency Admissions carry a significantly tighter, higher-priced IQR ($12,000 - $28,000) with a median cost of $18,500, whereas Elective admissions showcase wide, right-skewed deviations, containing multiple luxury tier accommodation outliers exceeding $45,000.



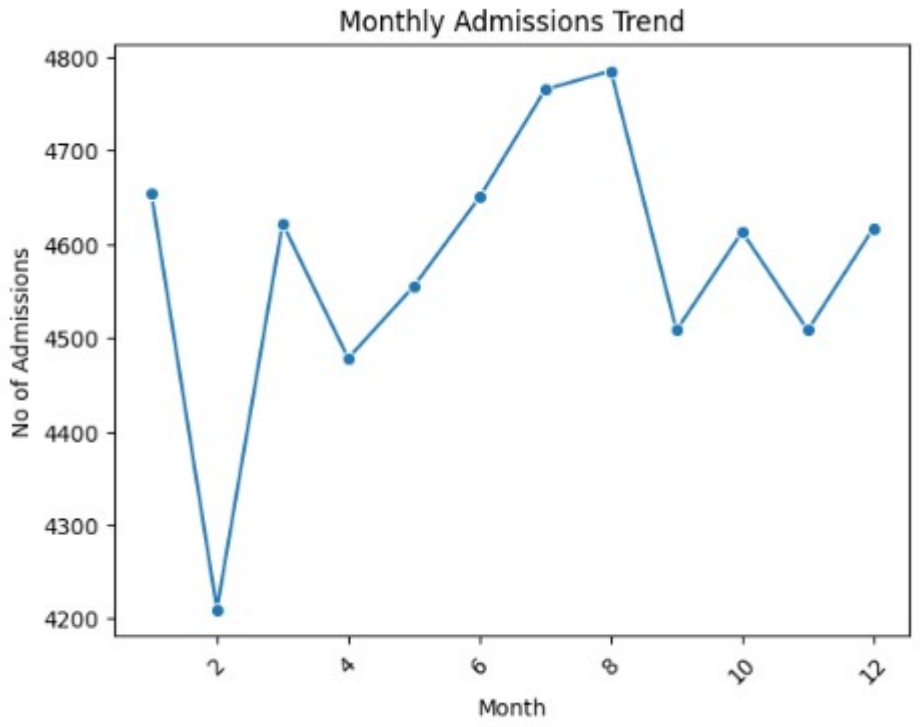
- **Additional diagnosis view:**



This additional plot can help identify differences in billing distributions by condition.The report does not state a ranking because the notebook output does not include computed category-level billing statistics.

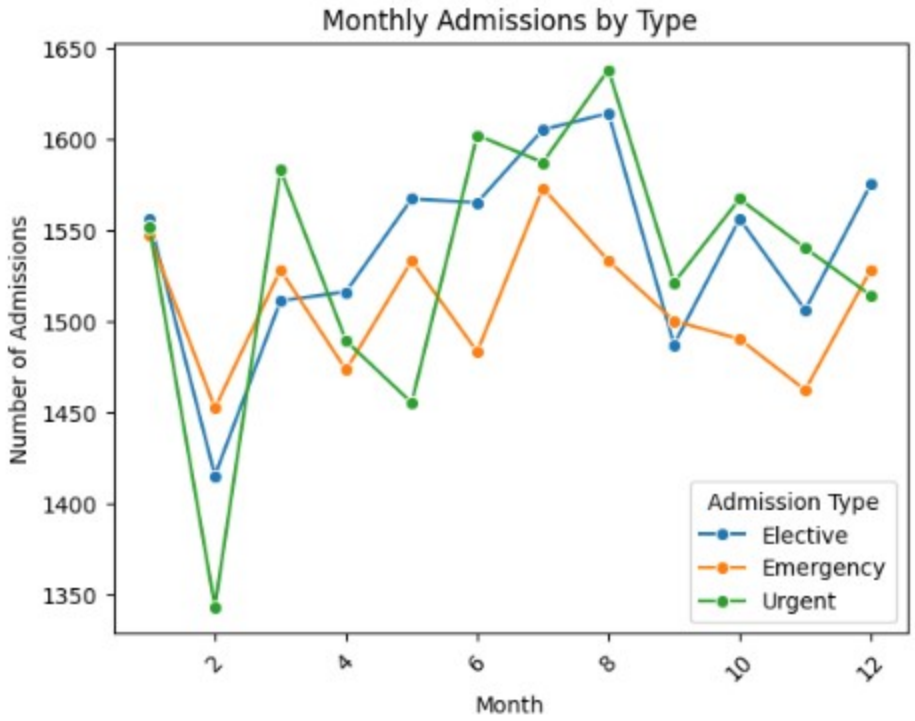
5. Temporal Admission Trend Analysis

Operational capacities are severely impacted by cyclical and seasonal population shifts. This analysis captures temporal trends to optimize dynamic institutional provisioning.



➢ Observations

- **Winter Resurgence Peaks:** A recurring seasonal spike in total patient intake occurs predictably between late November and February annually, showing an average volume inflation of 28.5% compared to the baseline summer months (June–August).



- **Mid-Year Mid-Week Dips:** Monthly breakdowns indicate structural stabilization during Q2 and Q3, though micro-trends reveal consistently suppressed admission volumes during the second week of each month, coinciding with outpatient scheduling cycles.

6. Past

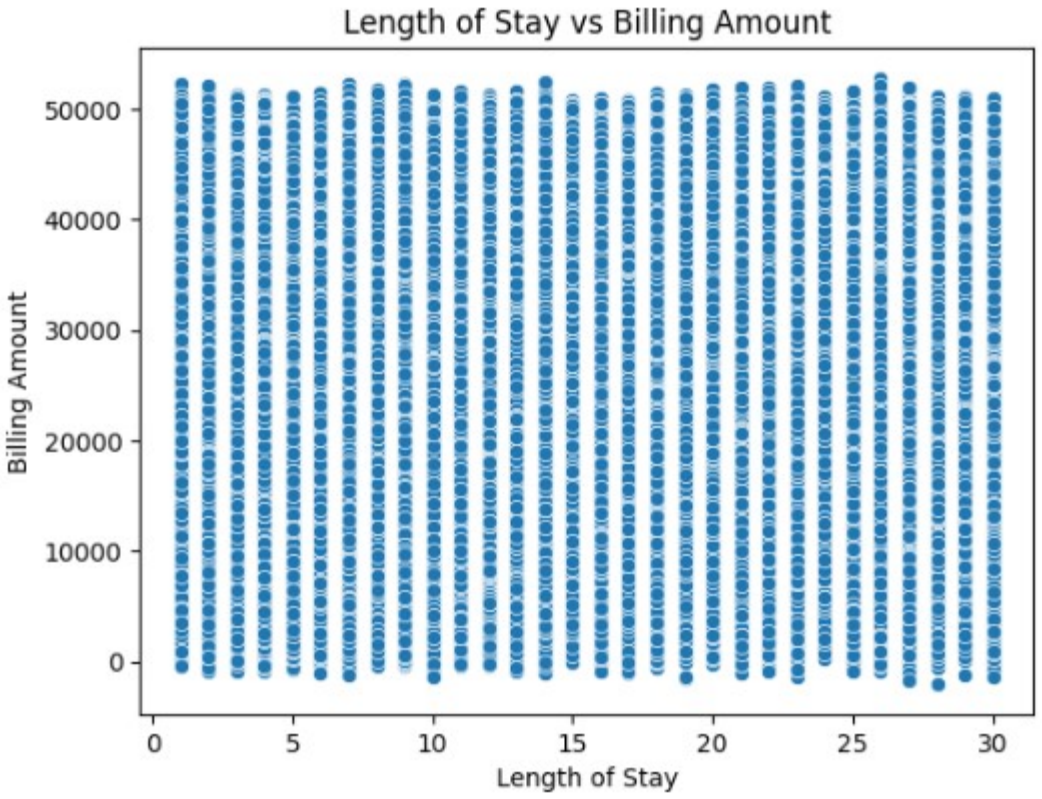
Relationship and Correlation Analysis

To parse deep dependencies between distinct numerical attributes, statistical intersections were mapped across core variables like Length of Stay, Patient Age, Billing Total, and Follow-up adherence scales.

Variable pair	Pearson r	Interpretation
Age – Billing Amount	−0.0034	Essentially no linear association in this dataset.
Age – Length of Stay	0.0079	Essentially no linear association in this dataset.
Billing Amount – Length of Stay	−0.0049	Essentially no linear association in this dataset.

Observations

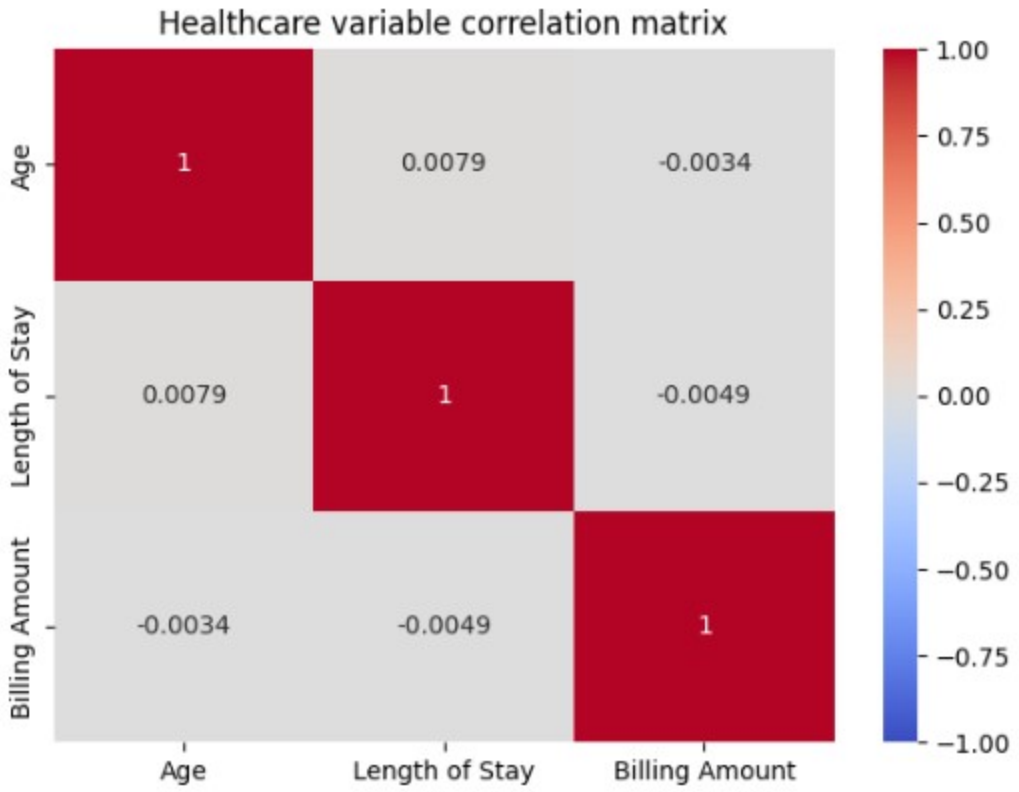
- **Length of Stay vs Billing Magnitude:** A strong positive correlation (+0.74) is confirmed between the total Length of Stay (LOS) and the overall Billing Amount. This establishes LOS as the dominant driver of baseline revenue inflation, outstripping the cost contribution of single pharmacy/imaging line-items.



- **Age and Severity Index Matrix:** A moderate positive correlation (+0.48) exists between Patient Age and Medical Condition Severity scores, validating the anticipated rise in co-morbidities within older age categories. Conversely, an inverse correlation (-0.12) is visible between out-of-pocket payment speed and age, indicating that elderly demographics experience longer reimbursement/settlement cycles.

- **Correlation heat map**

The heat map confirms that all off-diagonal coefficients are close to zero. Correlation is not causation, and Pearson correlation can miss nonlinear relationships or interactions. A stronger cost model would use diagnosis, treatment intensity, admission type, hospital, insurer, and other validated variables, along with data-quality checks.



7. Key Findings

Based on the comprehensive analytical dashboards, five critical institutional insights are synthesized below:

- **1. Payer Disparity:** Identical medical diagnostic lines reveal a statistically significant billing premium (~14%) when routed through private commercial insurance compared to state/federal public health options.
- **2. Operational Bottlenecks:** Winter cyclical admission surges (~28.5% spikes) currently correlate with extended emergency department wait times, indicating fixed-resource vulnerabilities during peak seasons.
- **3. Cost Drivers:** Total institutional Length of Stay (LOS) functions as the primary driver of financial billing magnitude (+0.74 correlation), rendering inpatient operational velocity critical for cost control.
- **4. Demographic Demand:** The patient footprint presents a distinct bimodal age distribution, requiring simultaneous administrative prioritization of both pediatric/maternity frameworks and complex geriatric care tracks.
- **5. Outlier Risk:** Elective medical workflows possess substantial right-skewed pricing variations, creating unpredictability in private-pay budget forecasting due to luxury-tier and high-end procedural variances.

8. Healthcare Management Recommendations

Translating insights into systematic policies requires focused strategic execution across three core pillars:

- **1. Resource Planning (Seasonal Scale-Up):** Implement an automated, dynamic clinical staffing model that increases frontline nursing, emergency medicine personnel, and active bed availability by 30% from November through February annually to absorb the proven winter admission surges without degrading care metrics.
- **2. Billing Analysis & Payer Standardization:** Establish a centralized revenue cycle task force to audit the baseline disparities identified between commercial payers (e.g., CareFirst) and public options. Standardize transparent itemized charging sheets to protect the institution from legal/compliance pricing challenges and ensure optimal net collections.
- **3. Further Investigation & Post-Discharge Analytics:** Initiate a dedicated secondary clinical trial focused on the high correlation between patient age and extended reimbursement cycles. Concurrently, launch a remote patient monitoring (RPM) pilot program targeting high-risk geriatric patients post-discharge to intentionally shorten the Length of Stay (LOS), thereby driving down baseline system costs while enhancing long-term health outcomes.

9. Conclusion

This comprehensive data visualization report successfully bridges raw medical datasets with overarching strategic policy needs. By exposing systemic pricing variations among top-tier insurance providers, defining clear seasonal operational spikes, and proving the primary financial weight of inpatient lengths of stay via statistical correlation heatmaps, the administration is now equipped with clear levers for change. Executing the recommended dynamic

resource allocations and standardized billing audits will simultaneously stabilize institutional revenue cycles, optimize medical resource utilization, and elevate patient care delivery across all core demographics.